

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

JENOPTIK AG (SwissOptic)

Frau Edith Memeti
Heinrich-Wild-Strasse
9435 Heerbrugg
Schweiz

Subject: Application as Process Engineer Metrology 100% (m/f/d)

Dear Ms. Memeti,

As a PhD physicist with deep expertise in optical metrology, I am applying for the position of Process Engineer at your Heerbrugg site. I am specifically seeking a role where I can apply my analytical skills to ensure your manufacturing processes operate at the highest level of precision.

During my PhD, my focus was on building and validating high-resolution optical measurement systems (interferometry, wavefront analysis). To ensure high process stability, I use LabVIEW and Python extensively for the automation of calibration procedures. Previously, at INTECOL S.A.S., I was responsible for the design of industrial inspection systems in compliance with industry standards (e.g., ISO 10110).

Fully integrated into the DACH work culture, I will be available for employment from May 2026. I look forward to sustainably optimizing development and qualification processes at SwissOptic/Jenoptik.

Sincerely,

Juan David Muñoz-Bolaños

Prozessingenieur Messtechnologie
Medizinische Universität Innsbruck

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Professional Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

TECHNICAL COMPETENCIES

Optical Metrology & System Design

End-to-end design and assembly of multiphoton/confocal microscopes and interferometric setups. Expertise in wavefront sensing and diffraction-limited system alignment.

Simulation & Tolerancing

Advanced use of ZEMAX OpticStudio for systematic tolerance analysis, sensitivity characterization of custom objectives, and Zernike-based wave-field simulations in Python.

Software & Automation

Expert proficiency in C++, Python (NumPy, Pandas, Scikit-learn, JAX), and LabVIEW/FPGA for real-time hardware control, detector calibration, and automation of calibration routines.

Industrial Vision & QA

Hands-on experience with Halcon and Cognex VisionPro for production-line inspection. Proficient in MSA (Gage R&R) and designing custom illumination concepts (Bright-field/Dark-field).

Standards & Compliance

Strong awareness of ISO 10110 and ISO 9001 standards. Experienced in managing EH&S compliance and maintaining technical documentation for high-power laser environments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **Metrology System Build:** Designed and automated a custom wavefront sensing and phase retrieval system from scratch; wrote complete control stacks in C++/Python to interface cameras with deformable mirrors.
- **Optical Characterization:** Performed rigorous ZEMAX tolerance analysis to characterize aberration sensitivity in custom-built microscope objectives, validated via experimental interferometry.
- **R&D Project Leadership:** Led the hardware-software integration of a multiphoton microscope system, achieving diffraction-limited performance while managing vendor qualifications (Thorlabs, Edmund Optics).
- **Lab Operations:** Primary operator for a high-power laser lab, managing safety compliance (EH&S), equipment maintenance logs, and technical spec-sheet documentation.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Spectrometer Prototyping:** Designed and built a 1064 nm fiber-probe Raman spectrometer for non-invasive clinical use; handled end-to-end optical layout, dispersive optic alignment, and detector calibration.
- **Data Science & ML:** Applied supervised Machine Learning to hyperspectral photonic data using custom Python packages, achieving high-accuracy tissue classification and accelerating experimental validation loops.

- **Interdisciplinary Research:** Collaborated on clinical spectroscopy projects, resulting in two peer-reviewed publications in Applied Sciences and ACS Photonics.

INTECOL S.A.S.
Junior Machine Vision System Engineer

Medellín, Colombia
Mar 2018 – Mar 2019

- **Industrial Vision Pipelines:** Engineered Halcon pipelines for fast-moving production lines (glass bottle inspection), implementing custom illumination concepts to solve liquid level and label printing quality issues.
- **System Commissioning:** Successfully deployed automated inspection solutions for a car subdivision of Renault, utilizing Cognex smart cameras and custom illumination methods for robotic part recognition and flame-surface treatment.
- **Process Reliability:** Conducted feasibility studies and Gage R&R assessments to ensure measurement traceability and process stability under industrial conditions.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for top Colombian professionals.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

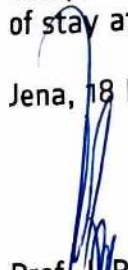
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

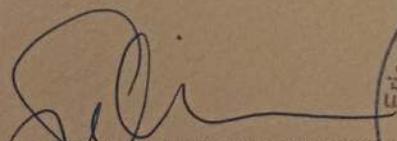
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Dear Family

[illegible]

